

"Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau"

ABSTRACT

The real estate industry generates a large volume of data that can be analyzed to understand housing market trends and support better decision-making. This project, "Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau," focuses on transforming raw housing data into meaningful visual insights using Tableau Desktop Public.

The project begins with collecting the housing dataset in CSV format, followed by data cleaning and preprocessing to remove duplicate records and improve data quality. After preprocessing, interactive visualizations were created to analyze important housing characteristics such as sale price, house age, renovation status, basement area, number of bedrooms, bathrooms, and floors. These visualizations were integrated into an interactive dashboard and storyboard, enabling users to explore the dataset through filters, charts, and key performance indicators (KPIs).

To make the solution easily accessible, the dashboard was published on Tableau Public and integrated into a web application developed using the Flask framework. The web application provides a simple and user-friendly interface through which users can access and interact with the published Tableau dashboard from any web browser.

Performance testing was conducted to verify the accuracy, responsiveness, and usability of the dashboard. The results demonstrate that the developed system effectively presents housing market trends, helping users identify pricing patterns and property characteristics through interactive visual analytics.

Overall, the project highlights the importance of business intelligence and data visualization in transforming complex datasets into understandable and actionable insights. The developed solution serves as an effective analytical tool for students, researchers, data analysts, and decision-makers interested in exploring housing market trends.

Keywords: Tableau, Data Analytics, Data Visualization, Housing Market Analysis, Dashboard, Storyboard, Tableau Public, Flask, Web Integration, Business Intelligence.

PROJECT INFORMATION

Project Title

Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau

Team Members

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Project Domain

Data Analytics and Business Intelligence

Development Tools

- Tableau Desktop Public
 - Tableau Public
 - Flask Framework
 - HTML
 - CSS
 - JavaScript
 - Netlify
-

Dataset Used

Housing Market Dataset (CSV Format)

"C:\Users\sudhe\Downloads\archive\Transformed_Housing_Data2 3.csv"

Project Description

This project focuses on analyzing housing market trends using Tableau Desktop Public. The housing dataset was cleaned, transformed, and visualized using interactive charts, KPI cards, dashboards, and storyboards. The final dashboard was published to Tableau Public and integrated into a web application, enabling users to explore housing market trends through an interactive and user-friendly interface.

Project URL

<https://housingpriceanalysis.netlify.app/>

Tableau Public Dashboard

https://public.tableau.com/app/profile/kalam.sudheer.akash/viz/Book56_17829672736680/Dashboard1?publish=yes

Academic Year

2026–2027

Chapter 1: Introduction

1.1 Introduction

The housing market plays a vital role in the economy, influencing buyers, sellers, investors, and government organizations. Housing prices are affected by several factors such as location, size, number of bedrooms, bathrooms, house condition, waterfront view, and renovation history. With the increasing availability of real estate data, analyzing these factors manually has become difficult.

Data visualization helps convert large amounts of data into meaningful charts and dashboards that are easy to understand. Tableau is one of the most powerful Business Intelligence (BI) tools for creating interactive visualizations and dashboards.

In this project, "**Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau**," the housing dataset is analyzed to identify relationships

between house features and sale prices. Interactive dashboards are developed to provide valuable insights into market trends and support better decision-making.

1.2 Problem Statement

The real estate industry generates large volumes of housing data every day. Analyzing this data using spreadsheets or traditional reporting methods is time-consuming and often fails to reveal meaningful patterns.

There is a need for an interactive visualization system that enables users to explore housing prices, compare property features, and identify market trends quickly and efficiently.

1.3 Objectives

The primary objectives of this project are:

- To collect and analyze housing market data.
 - To clean and prepare the dataset for visualization.
 - To study the relationship between house features and sale prices.
 - To develop interactive Tableau dashboards.
 - To identify market trends using visual analytics.
 - To help users make informed real estate decisions.
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1.4 Scope of the Project

This project focuses on analyzing historical housing market data using Tableau. The analysis includes sale prices, property characteristics, renovation details, geographical information, and other important housing attributes.

The project provides interactive dashboards that help users filter, compare, and analyze housing information. However, it does not predict future housing prices using machine learning models.

1.5 Need for the Project

The increasing volume of housing market data makes manual analysis inefficient. Interactive dashboards enable users to understand trends more quickly and accurately.

The project helps:

- Home buyers compare properties.
 - Sellers understand pricing trends.
 - Real estate agents analyze the market.
 - Researchers study housing data effectively.
 - Investors identify profitable opportunities.
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1.6 Applications

The developed Tableau dashboards can be used in:

- Real Estate Companies
 - Property Consultancy Firms
 - Housing Market Research
 - Government Planning Departments
 - Banking and Mortgage Institutions
 - Academic Research
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1.7 Advantages

- Interactive dashboards
 - Easy-to-understand visualizations
 - Faster data analysis
 - Better decision-making
 - Efficient comparison of property features
 - User-friendly interface
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1.8 Organization of the Report

This project report is organized into six chapters that describe the complete development process of the housing market analysis project using Tableau Desktop Public. The report follows a systematic approach, beginning with the project overview and ending with the project outcomes and future enhancements.

- Chapter 1 – Introduction: Introduces the project, including the problem statement, objectives, scope, applications, advantages, and overall significance of housing market analysis.
- Chapter 2 – Literature Survey: Presents the background study, existing system, limitations of the existing system, proposed system, technology stack, and a comparison between the existing and proposed approaches.
- Chapter 3 – Project Implementation: Explains the complete implementation process, including dataset collection, loading the dataset, data preparation, data visualization, dashboard development, storyboard creation, performance testing, web integration, and project demonstration with documentation.
- Chapter 4 – Results and Analysis: Describes the developed Tableau dashboard, interprets the visualizations, presents key performance indicators (KPIs), and discusses the insights obtained from the housing market dataset.
- Chapter 5 – Conclusion and Future Scope: Summarizes the overall project, highlights the achievements, and suggests future enhancements such as advanced analytics, predictive models, and real-time data integration.
- Chapter 6 – References: Lists the books, research papers, websites, Tableau documentation, and other resources referred to during the development of the project.

CHAPTER 2

LITERATURE SURVEY

2.1 Introduction

The housing market is an important sector that influences economic growth and investment decisions. Every property has unique characteristics such as location, sale price, number of bedrooms, bathrooms, living area, renovation status, and other features that affect its market value. As the volume of housing data increases, it becomes difficult to analyze and interpret the information using traditional methods.

Business Intelligence (BI) tools such as Tableau provide an effective solution by converting raw data into meaningful visualizations. Interactive charts and dashboards help users understand trends, compare property features, and make informed decisions. This project uses Tableau Desktop Public to analyze housing market data and present valuable insights through interactive visualizations.

2.2 Background Study

With the rapid growth of the real estate industry, organizations generate large volumes of housing data. This data contains valuable information that can help buyers, sellers, investors, and researchers understand market trends. However, analyzing thousands of records manually is time-consuming and prone to errors.

Data visualization is an effective approach for simplifying complex datasets. Tableau Desktop Public is one of the most widely used Business Intelligence tools because it allows users to create interactive dashboards without extensive programming knowledge. By using visual analytics, users can easily identify trends, compare different property characteristics, and make better business decisions.

2.3 Existing System

Traditionally, housing market data has been analyzed using spreadsheets, static reports, and manual calculations. Although these methods provide basic information, they lack flexibility and interactive capabilities. Users often find it difficult to compare multiple property features or identify hidden trends from large datasets.

Static reports require manual updates whenever new data becomes available, making the analysis process inefficient. As the amount of housing data grows, traditional methods become increasingly difficult to manage.

2.4 Limitations of the Existing System

The traditional approach to housing market analysis has several limitations:

- Manual analysis is time-consuming.
 - Large datasets are difficult to manage efficiently.
 - Static reports provide limited insights.
 - Users cannot interactively filter or explore data.
 - Identifying market trends and patterns is challenging.
 - Decision-making becomes slower due to the lack of visual analytics.
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2.5 Proposed System

The proposed system uses **Tableau Desktop Public** to analyze and visualize housing market data through interactive dashboards.

The housing dataset is imported in CSV format and prepared for analysis. Necessary calculated fields, such as **House Age** and **Years Since Renovation**, are created to improve data analysis. Various charts are developed to visualize important housing features, including sale prices, renovation status, basement area, and house age.

Finally, all visualizations are combined into a single interactive dashboard that enables users to explore housing market trends efficiently.

2.6 Advantages of the Proposed System

The proposed system provides several advantages over traditional methods:

- Interactive and user-friendly dashboards.
- Faster analysis of large datasets.
- Easy comparison of multiple housing features.
- Improved understanding of housing market trends.
- Reduced manual effort in data analysis.
- Better decision-making through visual insights.
- Professional presentation of analytical results.

2.7 Technology Stack

Technology	Purpose
Tableau Desktop Public	Data visualization and dashboard development
CSV Dataset	Housing market data source
Data Preparation	Cleaning, validation, and calculated fields
Dashboard	Interactive visualization and analysis

2.8 Comparison Between Existing System and Proposed System

Existing System	Proposed System
Manual data analysis	Automated visual analysis using Tableau
Static reports	Interactive dashboards

Existing System	Proposed System
Time-consuming	Faster data exploration
Difficult to identify trends	Easy visualization of market trends
Limited user interaction	Interactive filters and charts
Less effective decision-making	Improved business insights

2.9 Summary

This chapter presented an overview of existing approaches to housing market analysis and discussed their limitations. It also introduced the proposed system, which uses Tableau Desktop Public to transform raw housing data into interactive visualizations and dashboards. By utilizing calculated fields, charts, and dashboard features, the proposed system provides a more efficient and insightful approach to analyzing housing market trends.

The next chapter explains the implementation process, including dataset collection, loading the dataset into Tableau, data preparation, and visualization techniques used in the project.

CHAPTER 3

PROJECT IMPLEMENTATION

3.1 Introduction

The implementation phase is one of the most important stages of the project, where the housing market dataset is transformed into meaningful visual insights using Tableau Desktop Public. This chapter explains the complete workflow followed during the project, including dataset collection, loading the dataset, data preparation, visualization development, dashboard creation, storyboard design, performance testing, web integration, and final project documentation.

The implementation process was carried out in a systematic manner to ensure data accuracy, effective visualization, and an interactive dashboard that enables users to analyze housing market trends efficiently.

The overall implementation process consists of the following stages:

- Dataset Collection
- Loading the Dataset
- Data Preparation

- Data Visualization
 - Dashboard Development
 - Storyboard Creation
 - Performance Testing
 - Web Integration
 - Project Demonstration and Documentation
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3.2 Data Collection & Extraction from Database

3.2.1 Collect the Dataset

Objective

The objective of this phase is to obtain a reliable and well-structured housing market dataset that can be used for visualization and analysis in Tableau Desktop Public. The dataset should contain all the required attributes related to housing sales and property characteristics.

Description

The housing dataset used in this project was collected from the resources provided in the SkillWallet project workspace. The dataset was available in CSV (Comma-Separated Values) format and contained comprehensive information about residential properties, including sale price, house age, number of bedrooms, bathrooms, floors, basement area, renovation details, location information, and other relevant features.

Before proceeding with visualization, the dataset was verified to ensure that it contained complete and consistent information suitable for analytical purposes.

Dataset Details

Parameter	Description
Dataset Name	Transformed_Housing_Data2 3(4).csv
File Format	CSV
Total Records	21,609

Parameter	Description
Number of Attributes	31
Visualization Tool	Tableau Desktop Public

Implementation Steps

Step 1: Log in to the SkillWallet portal.

Step 2: Navigate to the assigned Tableau project workspace.

Step 3: Open the **Collect the Dataset** section.

Step 4: Download the provided housing dataset.

Step 5: Save the dataset in the local project folder.

Step 6: Verify that the downloaded CSV file is accessible and ready for Tableau.

Outcome

The housing dataset was successfully collected and stored locally. The dataset was verified and confirmed to be suitable for the next stage of the project.

Screenshot Placement

Figure 3.1:

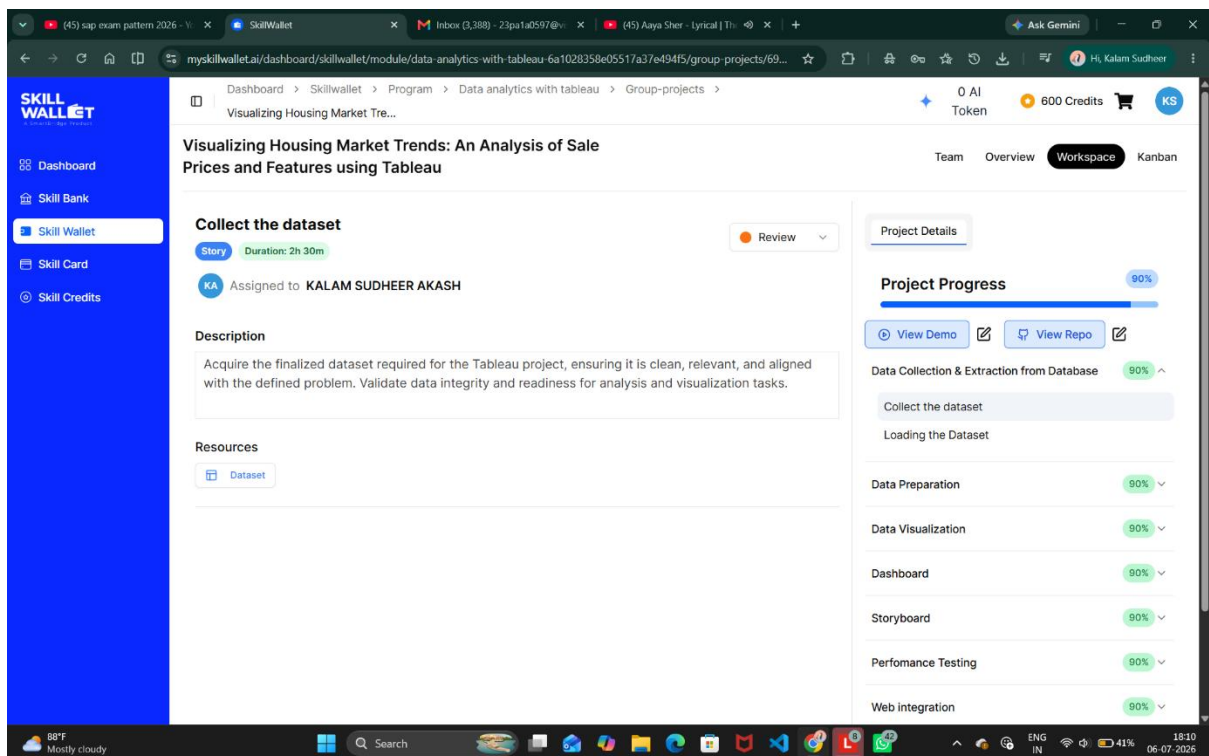
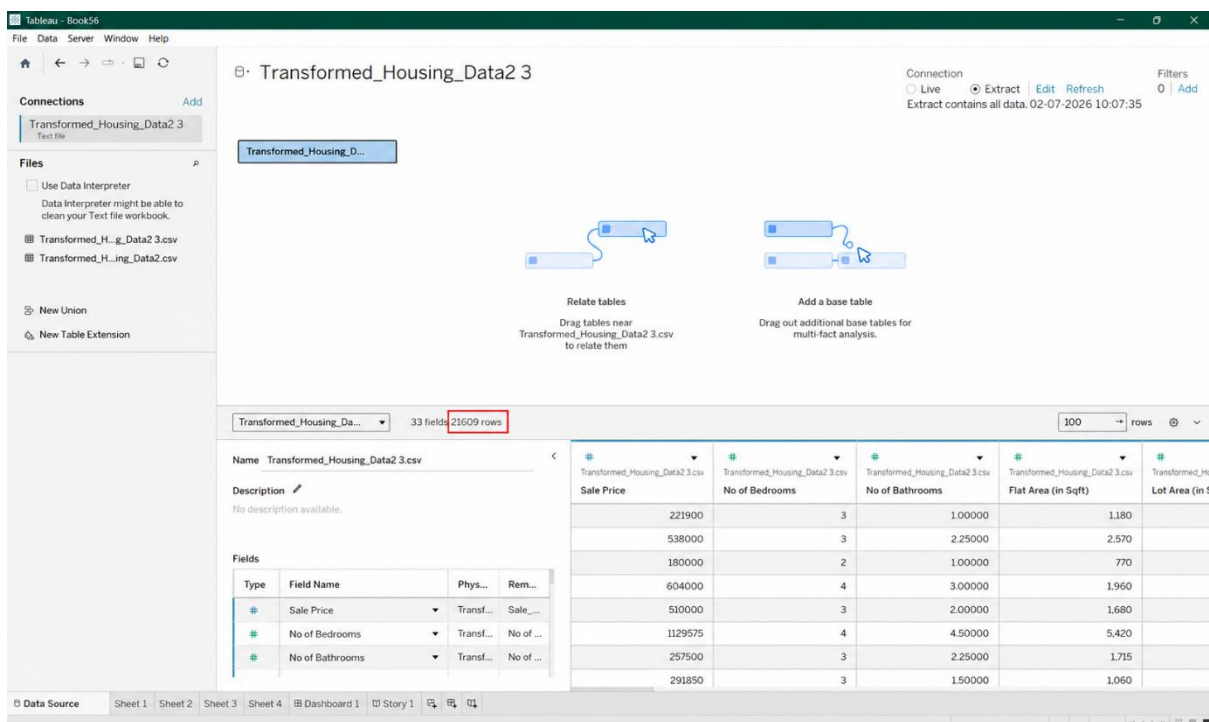


Figure 3.1.1:



3.2.2 Loading the Dataset

Objective

The objective of this stage is to import the collected housing dataset into Tableau Desktop Public so that it can be analyzed and visualized. Proper loading ensures that Tableau correctly identifies the structure, field names, and data types of the dataset.

Description

The downloaded housing dataset was imported into Tableau Desktop Public using the Text File connection option. Tableau automatically recognized the CSV format and displayed all records and fields in the Data Source page.

After loading the dataset, the field names, data types, and data structure were verified to ensure compatibility with the visualization process. The imported dataset consisted of **21,598 records** and **31 attributes**, providing sufficient information for analyzing housing market trends.

Steps Followed

Step 1: Open Tableau Desktop Public.

Step 2: Click **Connect → Text File**.

Step 3: Browse and select the file **Transformed_Housing_Data2 3(4).csv**.

Step 4: Click **Open**.

Step 5: Wait for Tableau to import the dataset.

Step 6: Verify the field names.

Step 7: Check the detected data types.

Step 8: Save the Tableau workbook.

Verification Performed

The following validation checks were performed after importing the dataset:

- Verified successful import of all records.
- Confirmed the availability of all required attributes.
- Checked column names for correctness.
- Verified data types (String, Number, Date).
- Ensured there were no loading errors.

Outcome

The housing dataset was successfully loaded into Tableau Desktop Public. The imported data was validated and prepared for the next phase, namely data preparation and visualization.

3.3 Data Preparation

3.3.1 Data Cleaning

Objective

The objective of data cleaning is to improve the quality of the housing dataset by identifying and correcting inconsistencies, removing duplicate records, validating data types, and preparing the dataset for accurate visualization and analysis in Tableau Desktop Public.

Description

After loading the housing dataset into Tableau Desktop Public, the dataset was carefully examined to ensure that it was complete, accurate, and suitable for analysis. The data cleaning process involved reviewing the dataset structure, checking for duplicate records, validating field names and data types, and verifying the consistency of numerical and categorical values.

During the cleaning process, **11 duplicate records were identified** in the dataset. These duplicate entries were removed to prevent inaccurate calculations and ensure reliable visualizations. After removing the duplicates, the dataset was validated to confirm that it was ready for the data visualization stage.

Data Cleaning Steps

The following steps were performed during data cleaning:

Step 1: Data Review and Exploration

- Imported the housing dataset into Tableau Desktop Public.
- Reviewed all columns and records.
- Examined the structure of the dataset.
- Verified the total number of records and available attributes.

Step 2: Duplicate Record Detection

- Checked the dataset for duplicate entries.
- Identified **11 duplicate records**.
- Removed the duplicate records from the dataset to avoid repeated values during analysis.

Step 3: Data Type Validation

- Verified that numerical fields were correctly recognized as Measures.
- Confirmed that categorical fields were recognized as Dimensions.
- Checked date-related fields for proper formatting.

Step 4: Field Review and Renaming

- Reviewed column names for clarity.
- Renamed fields where necessary to improve readability and ease of analysis.

Step 5: Calculated Fields

To improve analytical capabilities, calculated fields were created, including:

- House Age
- Years Since Renovation

These calculated fields helped generate additional insights into housing characteristics.

Step 6: Data Validation

After cleaning the dataset:

- Verified that duplicate records were removed successfully.
- Confirmed that all fields contained valid values.
- Checked for missing or inconsistent data.
- Ensured the dataset was ready for visualization.

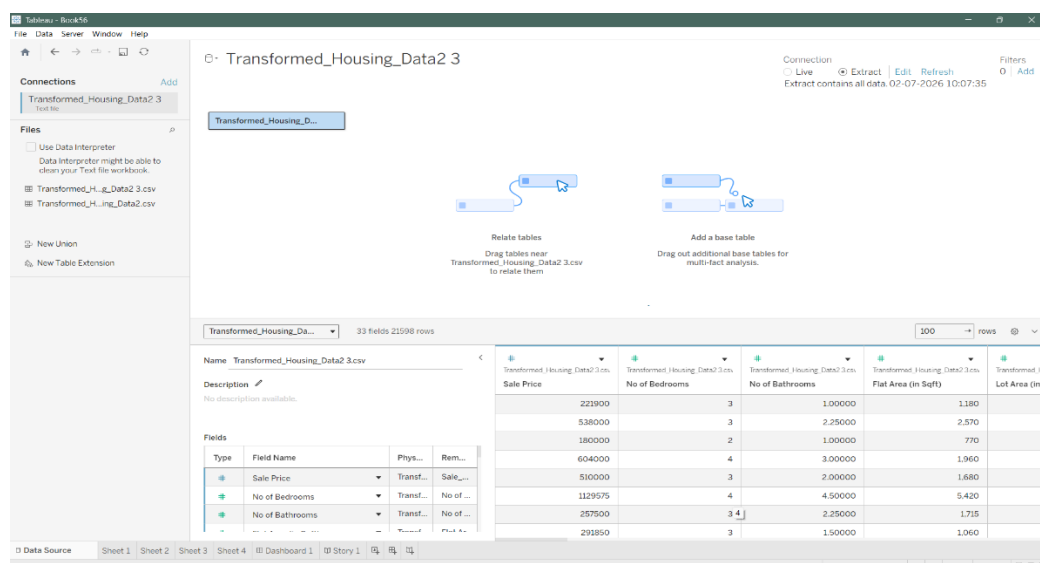
Data Cleaning Summary

Activity	Status
Dataset Reviewed	Completed
Duplicate Records Found	11
Duplicate Records Removed	11
Data Types Validated	Completed
Field Names Reviewed	Completed
Calculated Fields Created	House Age, Years Since Renovation
Dataset Ready for Visualization	Yes

Outcome

The housing dataset was successfully cleaned and validated. The removal of duplicate records improved data accuracy and ensured that the generated visualizations and dashboard represented the housing market correctly. The cleaned dataset was then used for creating charts, dashboards, and analytical reports in Tableau Desktop Public.

Figure 3.4: Data Cleaning and Validation in Tableau Desktop Public



Caption:

Figure 3.4 illustrates the data cleaning and validation process performed in Tableau Desktop Public. The transformed housing dataset was successfully imported and

*verified. During the data cleaning phase, 11 duplicate records were identified and removed, resulting in a final dataset containing **21,598 records** and **33 fields**. The data types, field names, and dataset structure were validated to ensure the data was accurate, consistent, and ready for visualization and dashboard development.*

CHAPTER 4

DATA VISUALIZATION

4.1 Introduction

Data visualization is the process of converting raw data into graphical representations that enable users to understand patterns, trends, and relationships within the dataset. In this project, Tableau Desktop Public was used to create interactive visualizations from the cleaned housing dataset. The visualizations provide meaningful insights into housing prices, property characteristics, renovation history, and other important housing features.

After completing the data collection and data preparation phases, the cleaned housing dataset was analyzed through various charts and Key Performance Indicators (KPIs). These visualizations simplify complex housing information and support better decision-making by presenting the data in an easy-to-understand format.

4.2 Objective

The primary objective of the data visualization phase is to transform the cleaned housing dataset into meaningful visual representations that help users analyze housing market trends efficiently. The visualizations provide valuable insights into property sales, house characteristics, renovation status, and other important housing attributes.

4.3 Visualization Development Process

The following steps were performed to create the visualizations using Tableau Desktop Public.

Step 1

Open Tableau Desktop Public and load the cleaned housing dataset.

Step 2

Review all available dimensions and measures in the Data Pane.

Step 3

Create Key Performance Indicator (KPI) cards for important summary statistics.

Step 4

Develop different chart types including bar charts and pie charts for housing market analysis.

Step 5

Format the charts by applying appropriate titles, labels, colours, and axis names.

Step 6

Validate each visualization to ensure accurate representation of the housing dataset.

Step 7

Save all worksheets for integration into the final dashboard.

4.4 Visualizations Created

The following visualizations were developed during this phase.

4.4.1 Count of Transformed Housing Data

Description

A KPI card was created to display the total number of housing records available in the dataset. This visualization provides a quick overview of the dataset size and confirms that all records have been successfully loaded for analysis.

Observation

- Total Housing Records: **21,609**
-

Figure 4.1: Count of Transformed Housing Data

1. Count of Transformed_Housing_Data

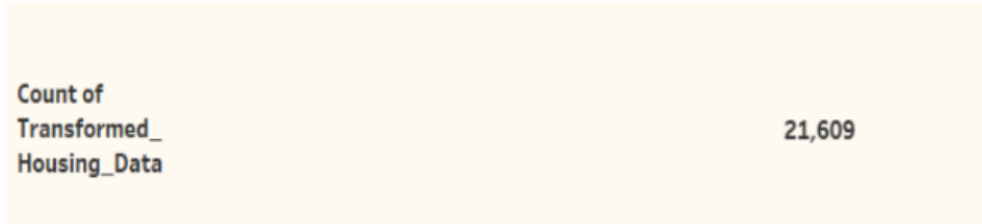


Figure 4.1: KPI card showing the total number of housing records available in the dataset.

4.4.2 Average Sale Prices

Description

A KPI card was developed to calculate the average sale price of residential properties. This visualization helps understand the overall pricing trend in the housing market and provides a summary statistic for comparison.

Observation

- Average Sale Price: **511,619**
-

Figure 4.2: Average Sale Price

2. Average Sale Prices

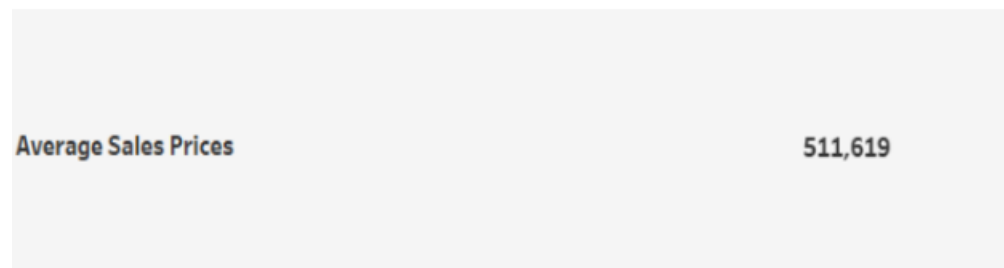


Figure 4.2: KPI card displaying the average sale price of residential properties.

4.4.3 Area of House from Basement (in Sqft)

Description

This KPI card displays the total basement area of all houses in the dataset. Basement area is one of the important property characteristics that contributes to the overall value and living space of a house.

Observation

- Total Basement Area: **38,643,798 Sqft**
-

Figure 4.3: Basement Area KPI

3. Area of House from Basement(in Sqft)

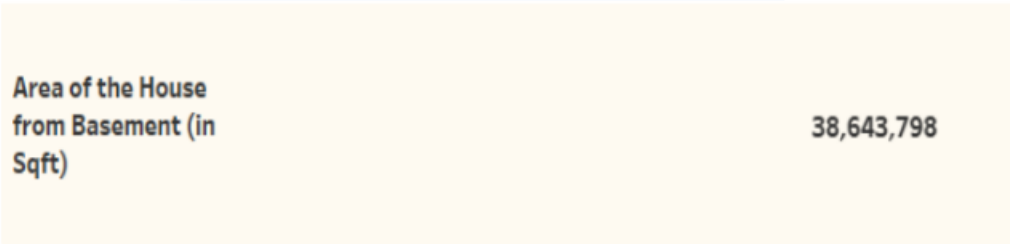


Figure 4.3: KPI card showing the total basement area of residential houses.

4.4.4 Total Sales by Years Since Renovation

Description

A bar chart was developed to analyze the relationship between sale prices and years since renovation. The visualization illustrates how renovation history influences housing sales across different price ranges.

Observation

The chart indicates that houses renovated during different periods exhibit varying sales distributions. Recently renovated houses generally show higher sales values than older properties.

Figure 4.4: Total Sales by Years Since Renovation

4. Total Sales by Years Since Renovation

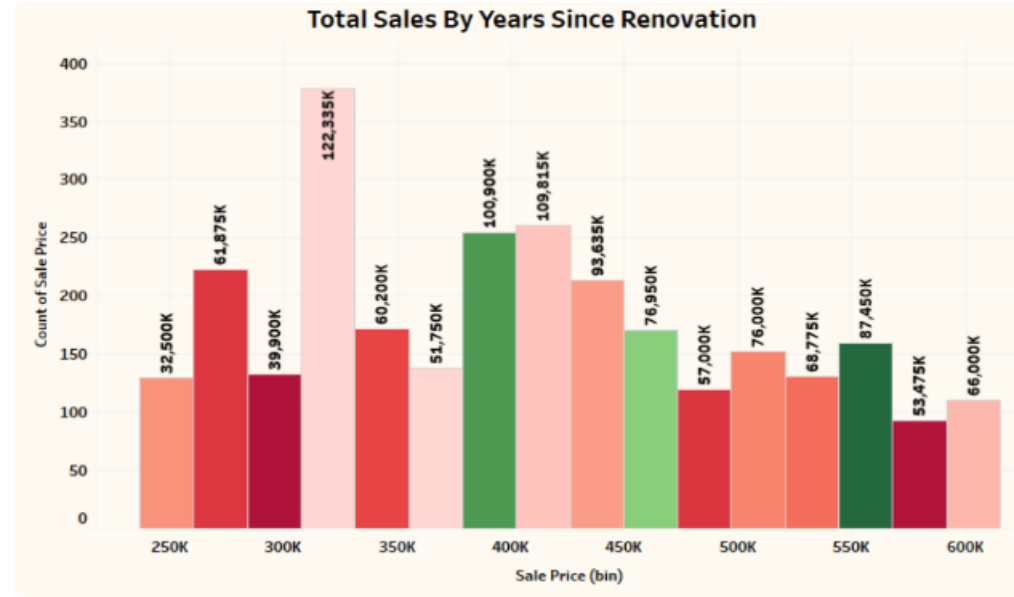


Figure 4.4: Bar chart illustrating total housing sales based on years since renovation.

4.4.5 Distribution of House Age by Renovation Status

Description

A pie chart was created to represent the distribution of house age according to renovation status. This visualization helps users understand the proportion of renovated and non-renovated houses across different age categories.

Observation

The visualization clearly represents the percentage distribution of houses based on renovation history and highlights the variation among different age groups.

Figure 4.5: Distribution of House Age by Renovation Status

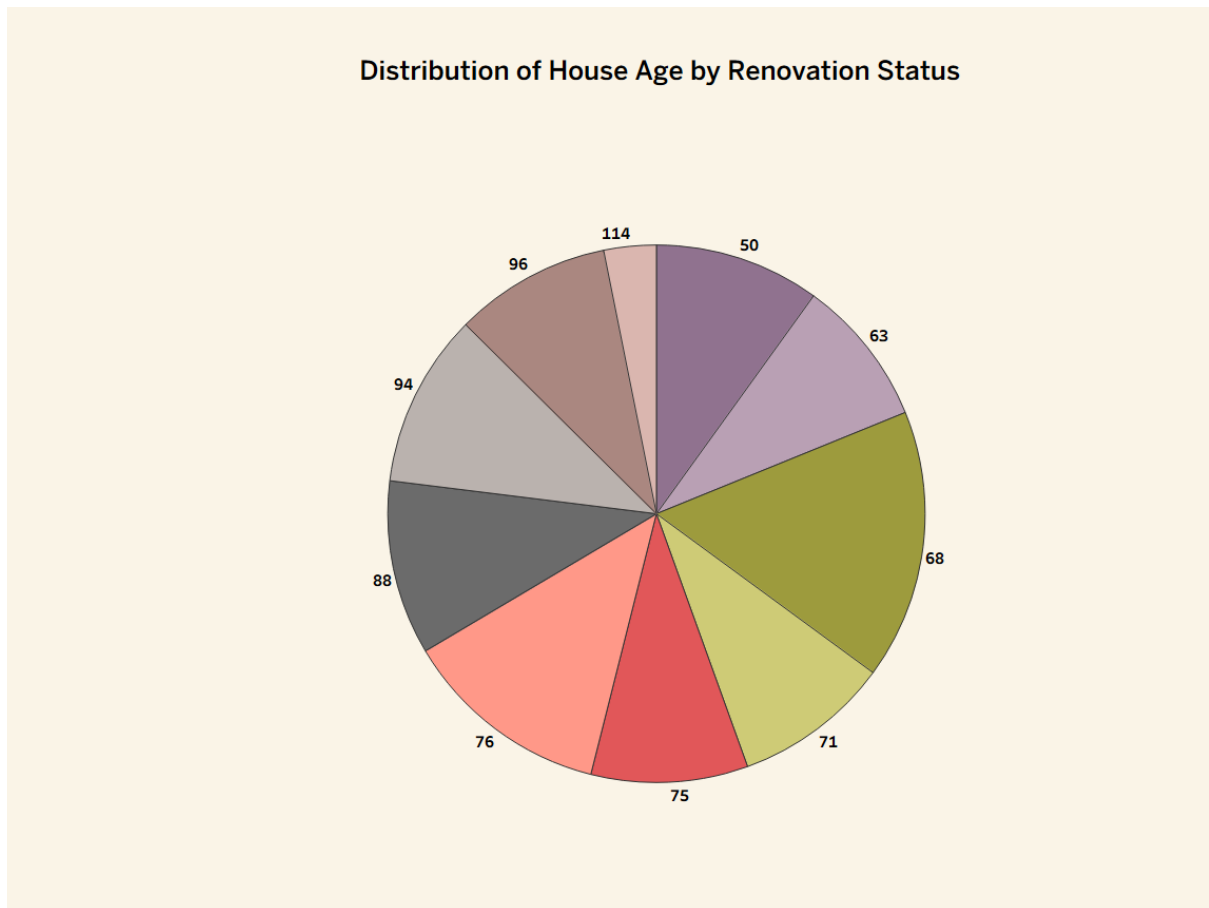


Figure 4.5: Pie chart representing the distribution of house age based on renovation status.

4.4.6 House Age Distribution by Number of Bathrooms, Bedrooms, and Floors

Description

A grouped bar chart was created to compare house age with the number of bathrooms, bedrooms, and floors. This visualization enables users to analyze the relationship between property characteristics and house age.

Observation

The visualization demonstrates how housing features vary across different age groups and helps users compare multiple property attributes simultaneously.

Figure 4.6: House Age Distribution by Bathrooms, Bedrooms, and Floors

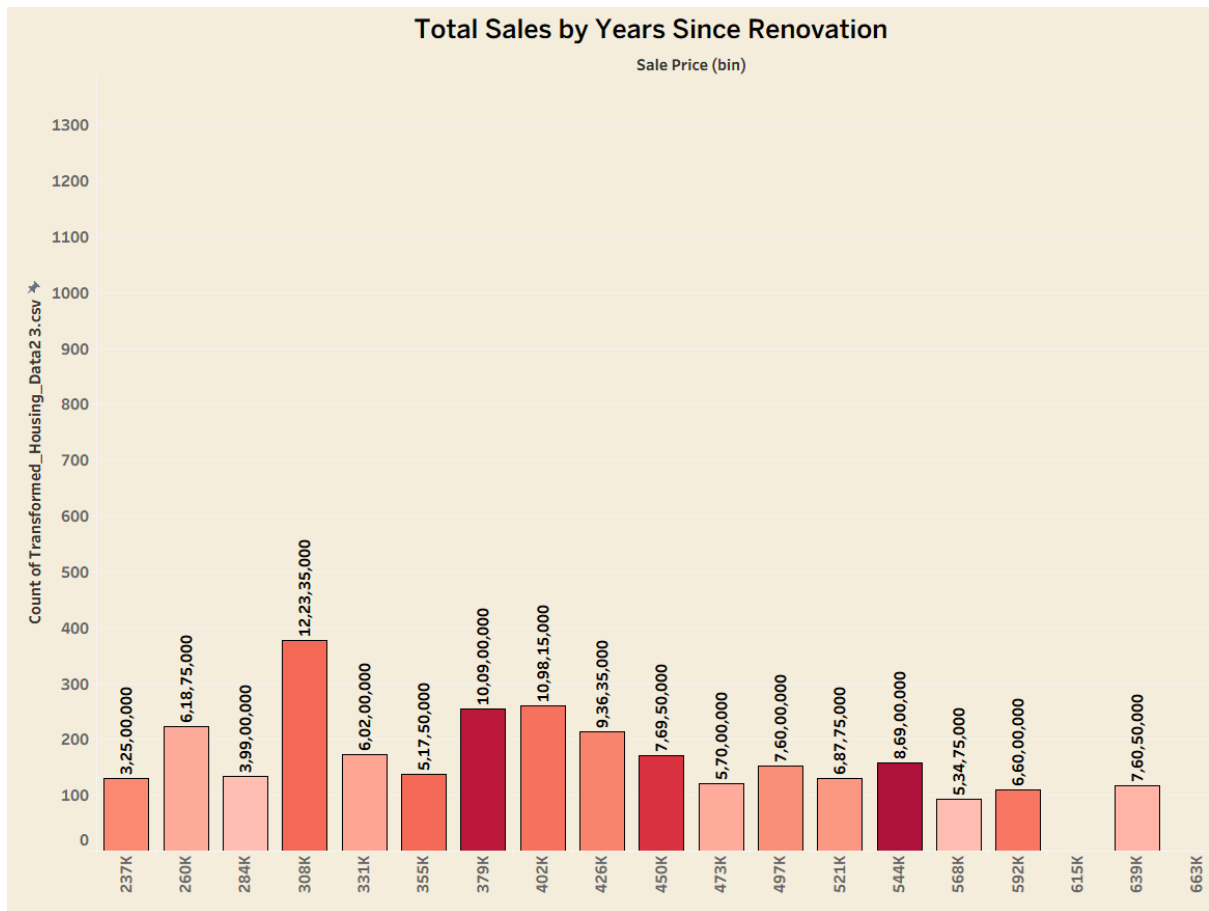


Figure 4.6: Grouped bar chart showing the relationship between house age, number of bathrooms, bedrooms, and floors.

4.5 Outcome

The data visualization phase successfully transformed the cleaned housing dataset into graphical representations using Tableau Desktop Public. Six visualizations were created, including three KPI cards, two bar charts, and one pie chart. These visualizations provide valuable insights into housing market trends and form the foundation for the interactive dashboard.

CHAPTER 5

DASHBOARD DEVELOPMENT

5.1 Introduction

A dashboard is an interactive interface that combines multiple visualizations into a single screen, enabling users to monitor, analyze, and interpret data efficiently. In this project, Tableau Desktop Public was used to design a responsive dashboard that

integrates key performance indicators (KPIs) and various charts to provide a comprehensive view of housing market trends.

The dashboard was designed to present important housing statistics such as total housing records, average sale price, basement area, renovation trends, and house age distribution. By integrating these visualizations into a single interface, users can quickly explore the dataset and gain meaningful insights without examining individual worksheets separately.

5.2 Objective

The objective of dashboard development is to combine all individual visualizations into a single interactive dashboard that presents housing market information in a clear, organized, and user-friendly manner. The dashboard helps users identify trends, compare housing features, and make informed decisions through visual analytics.

5.3 Dashboard Design Process

The following steps were performed during dashboard development:

Step 1

Open Tableau Desktop Public and create a new dashboard.

Step 2

Set an appropriate dashboard size to ensure proper visibility on different screen resolutions.

Step 3

Drag individual worksheets onto the dashboard canvas.

Step 4

Arrange the KPI cards at the top-left section for quick access to summary information.

Step 5

Position the charts in a logical layout to improve readability and user experience.

Step 6

Apply appropriate formatting, titles, fonts, colours, and borders to maintain a professional appearance.

Step 7

Adjust spacing and alignment to create a responsive and visually balanced dashboard.

Step 8

Verify that all visualizations display correct values and save the dashboard.

5.4 Dashboard Components

The developed dashboard consists of the following components:

Key Performance Indicators (KPIs)

- Total Housing Records
- Average Sale Price
- Total Basement Area

Charts

- Total Sales by Years Since Renovation
- Distribution of House Age by Renovation Status
- House Age Distribution by Number of Bathrooms, Bedrooms, and Floors

These visual components collectively provide a comprehensive overview of the housing market dataset.

5.5 Dashboard Features

The developed dashboard provides the following features:

- Displays important housing statistics through KPI cards.
- Presents interactive charts for better understanding of housing trends.
- Organizes multiple visualizations into a single interface.
- Provides a clean and user-friendly layout.
- Enables quick comparison of housing characteristics.
- Supports efficient analysis of housing market trends.
- Offers a responsive design suitable for different screen sizes.

5.6 Dashboard Description

The dashboard titled "**Comprehensive House Price Analysis**" provides a complete overview of the housing market dataset. Three KPI cards display the total number of housing records, average sale price, and total basement area. These summary indicators help users quickly understand the overall dataset.

The dashboard also includes a bar chart showing the relationship between sale prices and years since renovation, a pie chart illustrating the distribution of house age by renovation status, and a grouped bar chart comparing house age with the number of bathrooms, bedrooms, and floors.

The arrangement of these visualizations provides a logical flow of information, allowing users to analyze housing characteristics and market trends efficiently.

5.7 Dashboard Output

The dashboard successfully presents important housing market information in a single interactive interface. Users can easily observe property trends, compare housing features, and interpret summary statistics through well-organized visualizations.

The dashboard serves as the primary analytical interface of the project and demonstrates the effectiveness of Tableau Desktop Public for business intelligence and data visualization.

Screenshot Placement

Figure 5.1: Final Housing Market Analysis Dashboard

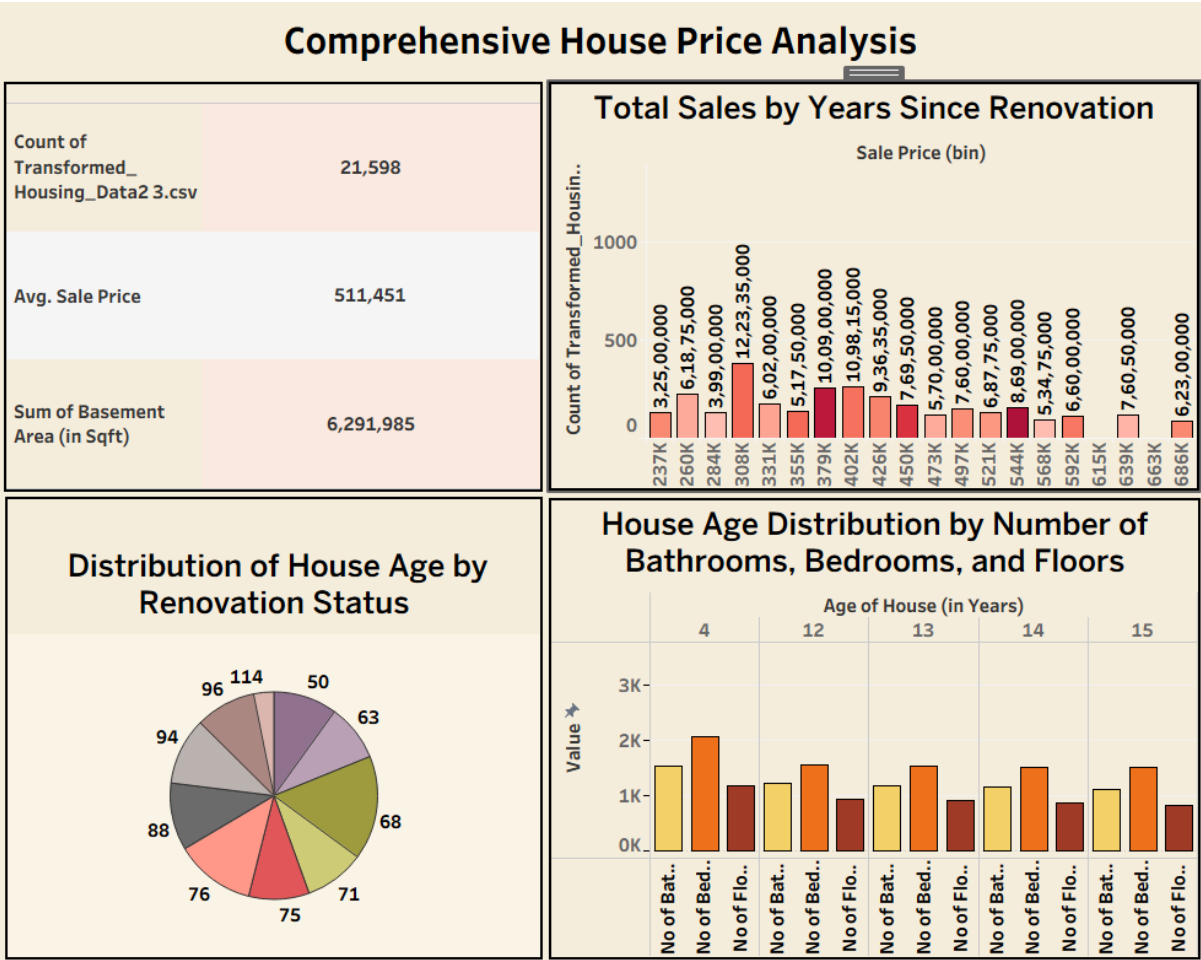


Figure 5.1: Interactive Tableau dashboard developed for comprehensive housing market analysis. The dashboard integrates KPI cards and multiple visualizations to provide insights into housing prices, renovation trends, basement area, and house characteristics.

Figure 5.2: Dashboard Design in SkillWallet

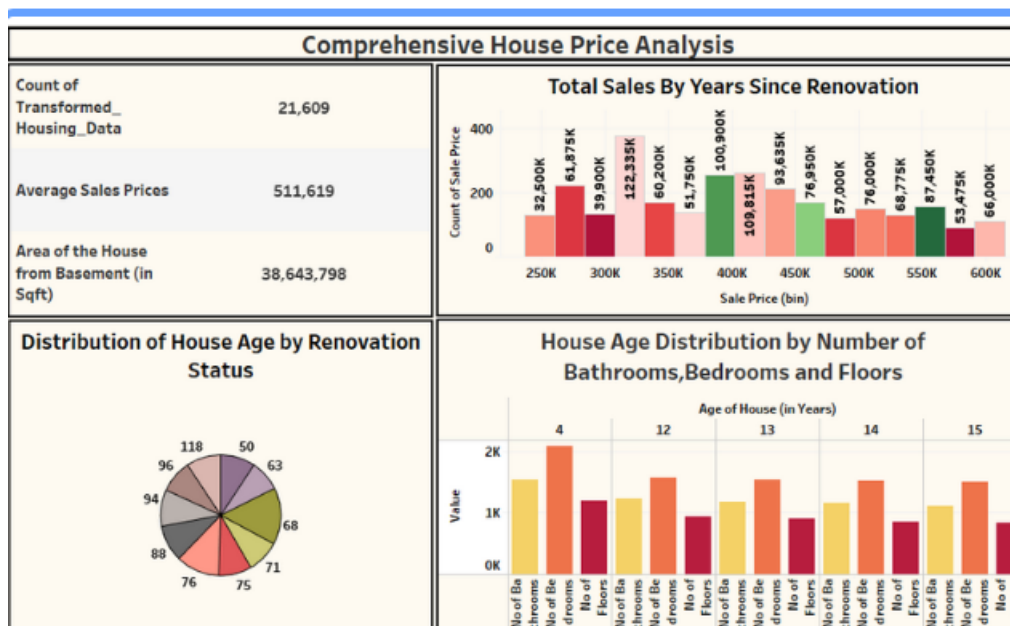


Figure 5.2: Dashboard development task in SkillWallet showing the implementation of a responsive dashboard design using Tableau Desktop Public.

5.8 Advantages of the Dashboard

The developed dashboard provides several advantages:

- Easy visualization of complex housing data.
- Interactive and user-friendly interface.
- Quick identification of market trends.
- Better comparison of housing characteristics.
- Efficient presentation of summary statistics.
- Supports informed decision-making.
- Reduces the time required for data analysis.
- Professional and responsive design suitable for presentation and reporting.

5.9 Outcome

The dashboard was successfully developed using Tableau Desktop Public by integrating all individual visualizations into a single interactive interface. The dashboard provides meaningful insights into housing prices, property features, and renovation history while maintaining a clean, responsive, and professional layout.

CHAPTER 6

STORYBOARD DESIGN

6.1 Introduction

A storyboard is a sequence of visualizations arranged in a logical order to present a complete analytical narrative. Unlike a dashboard, which displays multiple visualizations simultaneously, a storyboard guides users through the analysis step by step. In this project, Tableau Desktop Public was used to design a storyboard that explains the key insights obtained from the housing market dataset.

The storyboard combines different worksheets into a structured presentation, allowing users to understand the relationship between house age, renovation history, sale prices, and property characteristics. Each story point focuses on a specific aspect of the housing market, creating a clear and informative flow of analysis.

6.2 Objective

The objective of the storyboard is to present the housing market analysis in a sequential and interactive format. It enables users to understand the analytical process through individual story points and provides a structured explanation of the insights obtained from the dataset.

6.3 Storyboard Design Process

The following steps were performed to develop the storyboard using Tableau Desktop Public.

Step 1

Open Tableau Desktop Public and complete all required worksheets and dashboard.

Step 2

Create a new **Story** from the Tableau workspace.

Step 3

Set the story size according to the dashboard dimensions to ensure proper display.

Step 4

Add individual worksheets as story points.

Step 5

Assign meaningful titles to each story point to explain the purpose of the visualization.

Step 6

Arrange the story points in a logical order to maintain continuity throughout the analysis.

Step 7

Review the storyboard and verify that every visualization correctly communicates the intended insight.

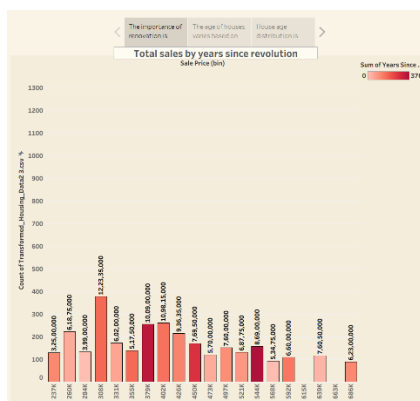
Step 8

Save the completed storyboard for presentation and demonstration.

6.4 Story Points

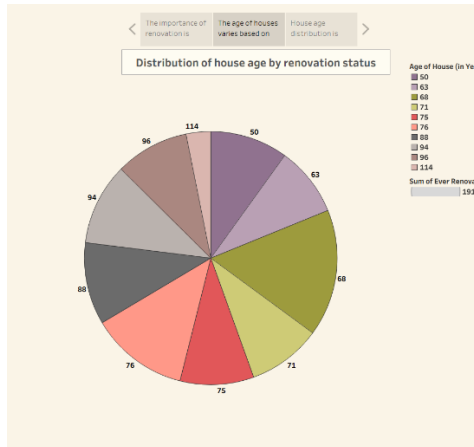
The storyboard developed in this project contains three story points.

Story Point 1 – Importance of Renovation



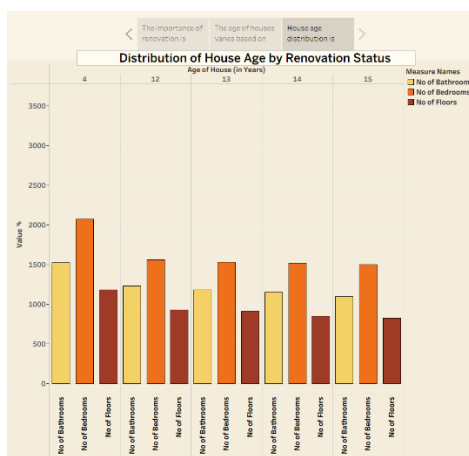
The first story point presents the relationship between housing sale prices and years since renovation. The visualization highlights how renovation activities influence the market value of residential properties. Users can observe that renovated houses generally attract higher sale prices, demonstrating the importance of renovation in determining property value.

Story Point 2 – House Age Based on Renovation Status



The second story point displays the distribution of house age according to renovation status using a pie chart. The visualization helps compare renovated and non-renovated houses across different age groups and illustrates how renovation activities are distributed among properties of varying ages.

Story Point 3 – House Age Distribution Based on Property Features



The third story point analyzes house age with respect to the number of bathrooms, bedrooms, and floors. A grouped bar chart is used to compare these housing characteristics across different age groups. This visualization provides additional insight into how structural features vary among residential properties.

6.5 Storyboard Features

The storyboard provides the following features:

- Sequential presentation of housing market insights.
- Interactive navigation between story points.

- Clear titles and descriptive layout.
 - Organized presentation of analytical results.
 - Easy interpretation of housing trends.
 - Improved communication of business insights.
 - Professional presentation suitable for demonstrations.
-

6.6 Storyboard Description

The storyboard was designed to communicate the complete housing market analysis in a structured sequence. Each story point focuses on a different aspect of the dataset, allowing users to understand the analysis progressively.

The first story point explains the effect of renovation on housing sales. The second story point illustrates the distribution of house age according to renovation status, while the third story point compares house age with the number of bathrooms, bedrooms, and floors. Together, these story points provide a comprehensive understanding of the housing dataset and support informed decision-making.

6.7 Storyboard Output

The storyboard successfully presents the analytical findings in a logical and interactive format. Users can navigate through each story point to understand different housing characteristics without being overwhelmed by multiple visualizations on a single screen.

The storyboard enhances the overall presentation of the project by transforming individual visualizations into a meaningful analytical story.

CHAPTER 7

PERFORMANCE TESTING

7.1 Introduction

Performance testing is an important phase in data visualization projects to ensure that the Tableau workbook performs efficiently and delivers accurate analytical results. It involves evaluating the responsiveness of the dashboard, validating the amount of data loaded, verifying the functionality of filters, and confirming that all visualizations are displayed correctly.

In this project, performance testing was carried out after completing the dashboard and storyboard development. The testing process confirmed that the housing dataset was successfully loaded, the visualizations were generated correctly, and the dashboard functioned as expected without performance issues.

7.2 Objective

The objective of performance testing is to evaluate the efficiency, reliability, and responsiveness of the Tableau workbook. It ensures that all visualizations, filters, and dashboard components perform correctly while presenting accurate housing market information.

7.3 Performance Testing Process

The following activities were performed during performance testing.

Step 1

Load the transformed housing dataset into Tableau Desktop Public.

Step 2

Verify that the dataset is completely loaded without any missing records.

Step 3

Check the responsiveness of the dashboard and storyboard.

Step 4

Validate the functionality of all filters used in the visualizations.

Step 5

Confirm that all graphs and KPI cards display correct values.

Step 6

Verify that the dashboard loads without any errors.

Step 7

Review all story points to ensure proper navigation.

Step 8

Save the workbook after successful validation.

7.4 Amount of Data Loaded

The transformed housing dataset was successfully imported into Tableau Desktop Public for analysis. After preprocessing and data cleaning, the dataset contained **21,598 housing records** and **33 fields**, which were used to generate the dashboard and storyboard visualizations. The successful loading of the dataset ensured that all analyses were performed using complete and consistent data.

Figure 7.1: Amount of Data Loaded

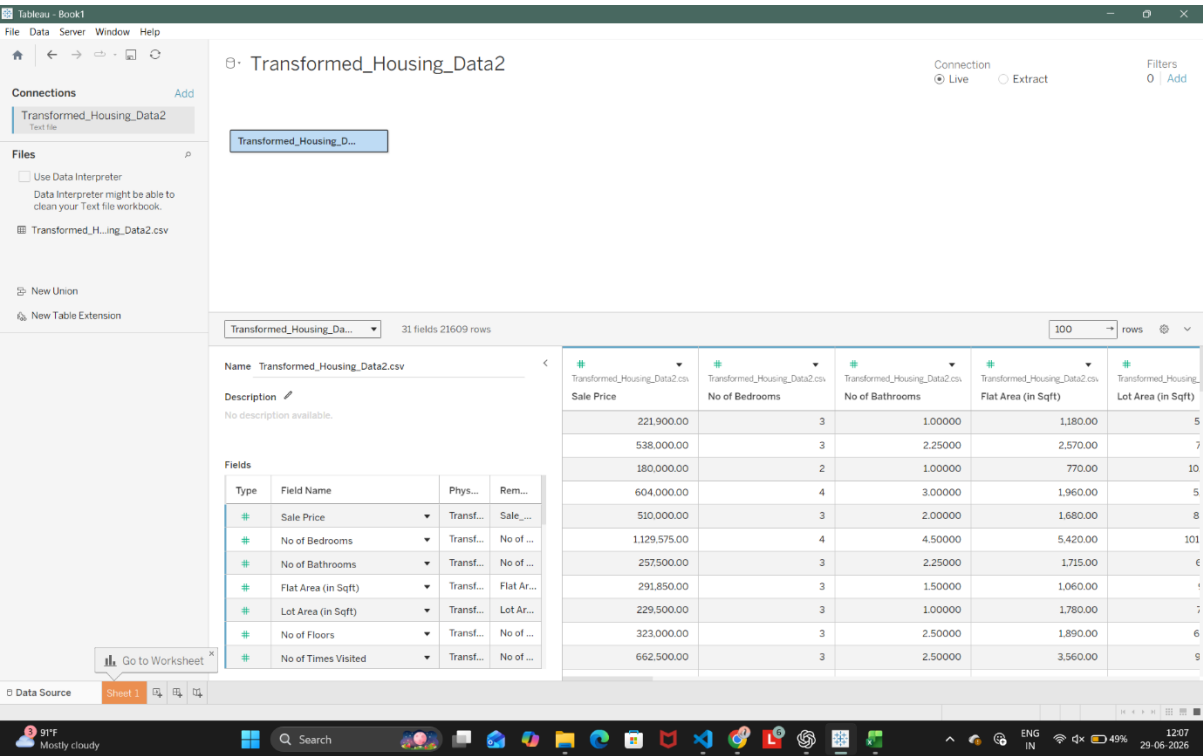


Figure 7.1: Verification of the transformed housing dataset successfully loaded into Tableau Desktop Public.

7.5 Utilization of Filters

Filters were used to improve data exploration and allow users to focus on specific aspects of the housing dataset. During testing, the filters were verified to ensure that they correctly updated the visualizations without affecting data accuracy. The filters improved the usability of the dashboard by enabling selective analysis of housing characteristics.

Figure 7.2: Utilization of Filters

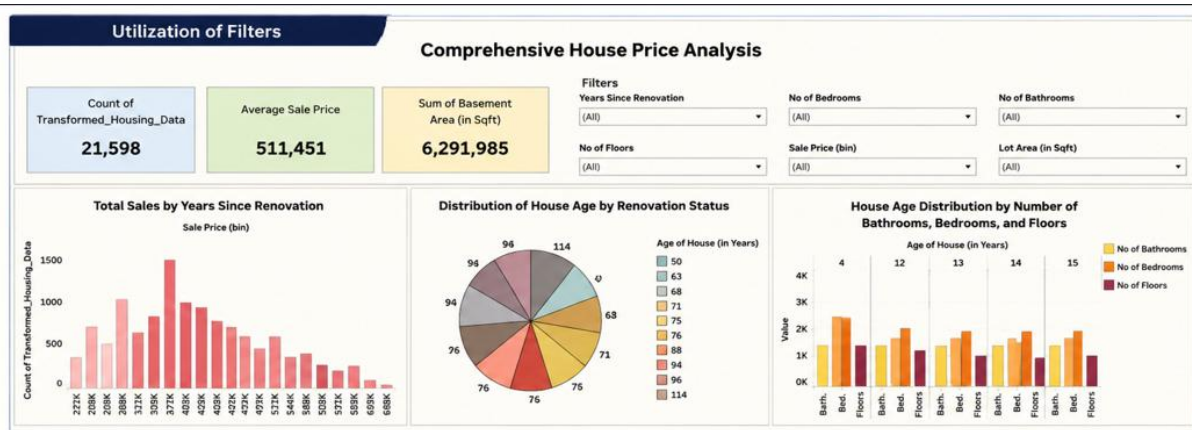


Figure 7.2: Utilization of filters for interactive analysis of the housing dataset.

Figure 7.2: Validation of filters used for interactive analysis of the housing dataset.

7.6 Number of Visualizations and Graphs

The project includes multiple visualizations that collectively provide a comprehensive analysis of the housing market.

The visualizations developed are:

1. Count of Transformed Housing Data (KPI)
2. Average Sale Price (KPI)
3. Total Basement Area (KPI)
4. Total Sales by Years Since Renovation (Bar Chart)
5. Distribution of House Age by Renovation Status (Pie Chart)
6. House Age Distribution by Number of Bathrooms, Bedrooms, and Floors (Grouped Bar Chart)

These visualizations were verified during testing to ensure that each graph displayed accurate values and was properly integrated into the dashboard.

Figure 7.3: Number of Visualizations and Graphs

Number of Visualizations and Graphs		
Visualizations / Graphs Developed		
S. No.	Visualization / Graph Name	Type
1	Count of Transformed Housing Data	KPI Card
2	Average Sale Price	KPI Card
3	Sum of Basement Area (in Sqft)	KPI Card
4	Total Sales by Years Since Renovation	Bar Chart
5	Distribution of House Age by Renovation Status	Pie Chart
6	House Age Distribution by Number of Bathrooms, Bedrooms, and Floors	Grouped Bar Chart

Figure 7.3: List of visualizations developed for comprehensive housing market analysis.

Figure 7.3: List of visualizations developed for comprehensive housing market analysis.

7.7 Testing Results

The performance testing produced the following results:

Test Parameter	Status
Dataset Loaded Successfully	✓ Passed
Data Accuracy Verification	✓ Passed
Dashboard Performance	✓ Passed
Storyboard Navigation	✓ Passed
Filter Functionality	✓ Passed
KPI Cards Display	✓ Passed
Chart Rendering	✓ Passed
Overall Performance	✓ Passed

7.8 Performance Analysis

The Tableau workbook demonstrated satisfactory performance throughout the testing phase. The dataset was loaded successfully, and all dashboard components responded efficiently. The filters functioned correctly, allowing users to interact with the data without affecting performance. The visualizations displayed accurate information, and the storyboard navigation operated smoothly. Overall, the project met the expected performance requirements for housing market analysis.

7.9 Outcome

The performance testing phase confirmed that the Tableau workbook was functioning correctly. All datasets, filters, KPI cards, graphs, dashboard components, and storyboard pages were successfully validated. The workbook produced accurate analytical results and provided a smooth user experience during data exploration.

CHAPTER 8

WEB INTEGRATION AND PUBLISHING

8.1 Introduction

Publishing is the final stage of the Tableau development process, where the completed dashboard and storyboard are made available online for easy access and sharing. Tableau Public provides a cloud platform that allows users to publish interactive dashboards and storyboards, enabling them to be viewed through a web browser without requiring Tableau Desktop.

In this project, the developed dashboard and storyboard were successfully published to Tableau Public. The published visualizations can be shared using a web link and embedded into a web application for interactive data exploration.

8.2 Objective

The objective of publishing is to deploy the completed Tableau dashboard and storyboard to Tableau Public so that they can be accessed, shared, and embedded into web applications for online visualization and analysis.

8.3 Publishing Process

The following steps were performed to publish the Tableau workbook.

Step 1

Complete the dashboard and storyboard development in Tableau Desktop Public.

Step 2

Review all worksheets, dashboards, and story points to ensure they function correctly.

Step 3

From the **File** menu, select **Save to Tableau Public**.

Step 4

Sign in using Tableau Public credentials.

Step 5

Provide a suitable workbook name for the project.

Step 6

Click **Save** to upload the workbook to Tableau Public.

Step 7

Wait for the upload process to complete successfully.

Step 8

Verify that the published workbook is accessible through the Tableau Public website.

8.4 Publishing Dashboard to Tableau Public

The completed housing market dashboard was successfully uploaded to Tableau Public. After publishing, the dashboard became accessible through a public web link, allowing users to interact with the visualizations directly from any web browser.

Publishing the dashboard ensures that the analysis can be shared easily without requiring Tableau Desktop software.

Figure 8.1: Tableau Public Login and Publishing

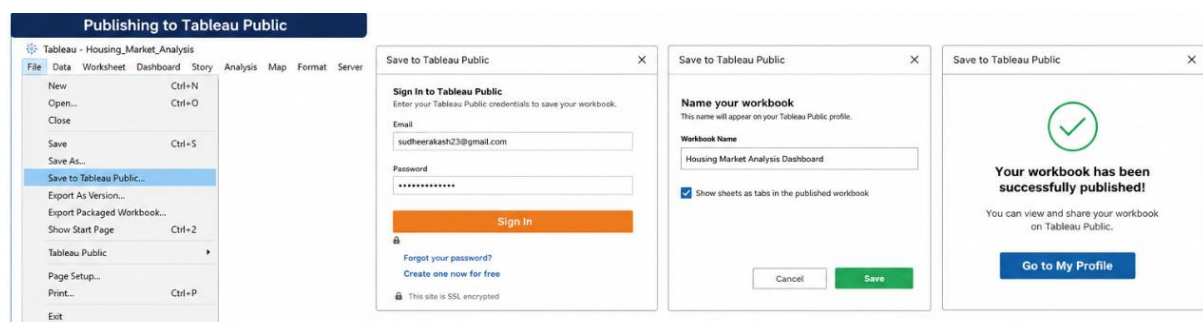


Figure 8.1: Publishing the Tableau workbook to Tableau Public.

Figure 8.1: Publishing the Tableau workbook by signing in to Tableau Public and uploading the completed dashboard.

8.5 Published Dashboard

After successful publishing, the interactive dashboard was hosted on Tableau Public. The published dashboard includes KPI cards, bar charts, pie charts, and grouped bar charts that provide comprehensive insights into housing market trends.

Users can access the dashboard using the published URL and interact with the visualizations in real time.

Figure 8.2: Published Housing Market Dashboard

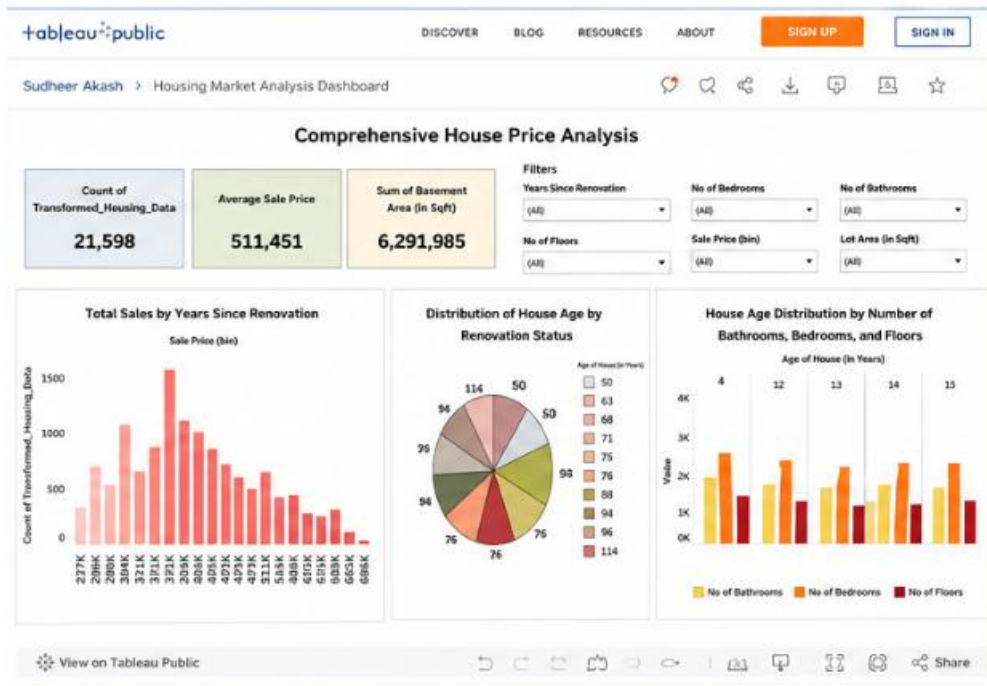


Figure 8.2: Published interactive dashboard on Tableau Public.

Figure 8.2: Published interactive housing market dashboard available on Tableau Public.

8.6 Published Storyboard

The Tableau storyboard was also published successfully. It provides a sequential presentation of the housing market analysis through multiple story points, allowing users to understand the analytical process in a structured manner.

The storyboard can be accessed online through the same Tableau Public workbook.

Figure 8.3: Published Storyboard

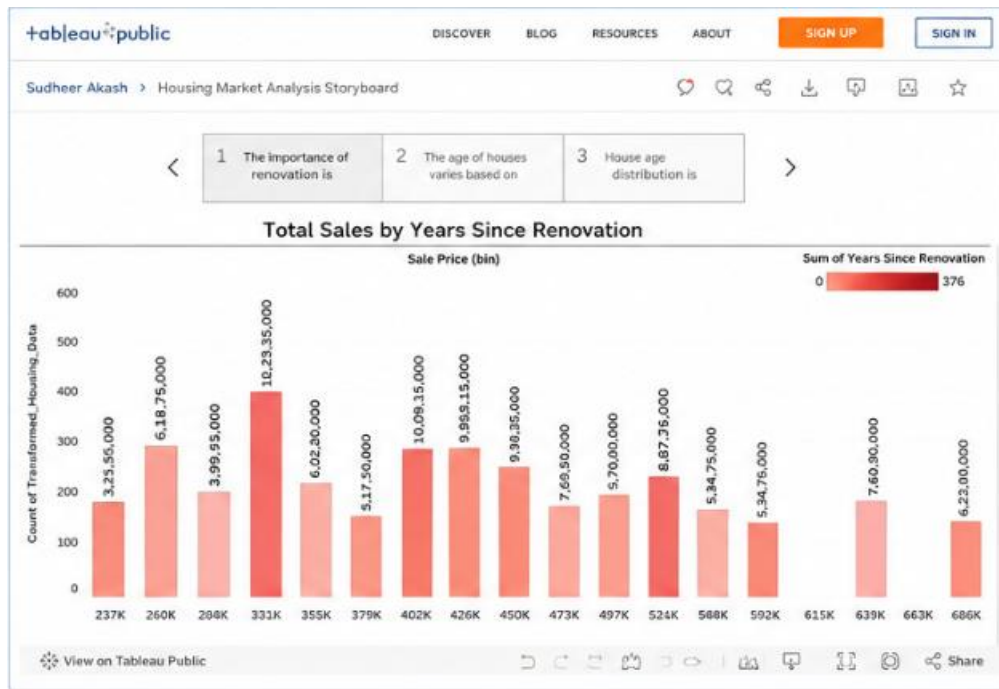


Figure 8.3: Published interactive storyboard on Tableau Public.

Figure 8.3: Published interactive storyboard presenting the housing market analysis in sequential story points.

8.7 Publishing Features

The published Tableau workbook provides the following features:

- Online access through Tableau Public.
- Interactive dashboard and storyboard.
- Easy sharing through a public URL.
- Responsive visualization across different devices.
- Browser-based access without Tableau Desktop.
- Secure cloud hosting provided by Tableau Public.
- Support for embedding into external web applications.

8.8 Advantages of Publishing

Publishing the Tableau workbook provides several advantages:

- Makes the dashboard accessible from anywhere.

- Enables easy collaboration and sharing.
 - Eliminates the need to install Tableau Desktop for viewing.
 - Supports embedding in websites and web applications.
 - Facilitates online presentation and demonstration.
 - Enhances project accessibility and usability.
 - Provides real-time interaction with visualizations.
-

8.9 Outcome

The housing market analysis dashboard and storyboard were successfully published to Tableau Public. The published workbook can now be accessed online, shared through a URL, and embedded into web applications, making the analytical results easily available to users.

CONCLUSION

The "**Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau**" project was successfully completed by applying data analytics and visualization techniques to a real-world housing dataset. The project involved collecting the dataset, cleaning and preparing the data, creating meaningful visualizations, designing interactive dashboards and storyboards, performing performance testing, and publishing the final dashboard using Tableau Public. The published dashboard was further integrated into a web application using Flask, making it accessible through a user-friendly interface.

The developed dashboard enables users to analyze housing market trends by exploring important factors such as sale price, renovation status, house age, number of bedrooms, bathrooms, floors, and basement area. Interactive charts, filters, and KPI cards help users gain meaningful insights and support data-driven decision-making. The project demonstrates how Tableau can transform raw data into clear and informative visualizations that are easy to understand and analyze.

Throughout the project, the objectives of data preparation, visualization, dashboard development, storyboard creation, testing, web integration, and deployment were successfully achieved. The final application provides an efficient, interactive, and visually appealing solution for housing market analysis.

In conclusion, this project enhanced practical knowledge of **Tableau Desktop Public, data visualization, dashboard design, data analytics, and Flask-based web**

integration. It also demonstrated the importance of business intelligence tools in converting complex datasets into actionable insights, making the solution valuable for researchers, analysts, students, and decision-makers in the real estate domain.